ECON 0100 | Vignette C1 | Externalities

Due in Recitation. Vignettes are a certificate of the work done together in Recitation.

Hogsmeade has recently seen a surge in shops selling Glittering Gum, a candy that makes one's words sparkle and glisten as one speaks. The Demand curve and Supply curve for Glittering Gum can be represented using the following relationships:

$$D : P_b = 30 - \frac{1}{2}Q_d \qquad S : P_s = 6 + \frac{1}{4}Q_s$$

After years of careful epidemiological analysis, a subcommittee of the Ministry tasked with improving the health of the wizarding community published a story in the Daily Profit establishing a link between the consumption of Glittering Gum and "Starlight Sneeze," a condition where non-consumers start sneezing out harmless but bothersome glitter. The story estimated this condition carries an inconvenience cost of 6 Coins per unit to those in Hogsmeade.

Use a graph and algebra to find equilibrium price, quantity, and DWL.

Equilibrium Price ($\hat{P}$): _____ Equilibrium Quantity ($\hat{Q}$): _____

Efficient Quantity (Q^*): _____ Deadweight Loss (DWL): _____